Ai Powered Document Processing System Using LangChain & Semantic Search

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Abstract

Managing vast volumes of data saved in PDF documents can be difficult, particularly when you're looking for precise answers fast. Simple keyword searches, which are frequently used by traditional systems, can miss the context of the query and yield results that are irrelevant. In order to address this issue, we developed a multi-PDF question-answering system that makes use of advanced artificial intelligence (AI) technology to comprehend and respond to queries based on the information contained in submitted PDFs. Using OpenAI embeddings to comprehend text, Pinecone to quickly store and retrieve information, and Groq (LLaMA 3) to generate replies, our system combines advanced technologies. The entire process, from uploading documents to producing answers, is managed by our system using LangChain. The technology divides a user-uploaded PDF into manageable chunks and examines the information. After that, it stores the data in a format that facilitates searching. The technology quickly finds the most relevant section of the document and gives a precise, understandable response when a user asks a question. This facilitates quick, intelligent, and effective searching through large PDFs

Our system's main features include semantic search, automated text extraction, and easy-to-use features like chat history logging and file uploads. Users can more easily and rapidly access information because to our system's proven ability to give accurate and useful replies. With the potential to help numerous sectors that depend on effective information retrieval, this initiative is a major advancement in the use of AI to enhance our interactions with documents.

Keywords- Multi-PDF Question answering, Artificial Intelligent, LangChain, Groq (LLaMA 3), OpenAI Embedding, MongoDB.

INTRODUCTION

In the current digital era, the amount of unstructured data has increased dramatically, especially in the form of PDF documents. PDFs are widely used for distributing and storing information, from commercial reports to study articles. But getting useful information out of these documents is still quite difficult. Traditional information retrieval techniques, including keyword-based searches, frequently produce answers that are insufficient or irrelevant because they are unable to understand the context of user queries. When

dealing with complicated, multi-part enquiries or huge document collections, this constraint is most noticeable. Consequently, there is an increasing demand for intelligent systems that can accurately and contextually respond to PDFs while processing, analysing, and retrieving information rapidly.

A Multi-PDF Question-Answering System Powered by large Language Models (LLMs), Semantic Search, and Retrieval-Augmented Generation (RAG) is proposed to overcome the difficulties of collecting relevant data from big PDF documents. The system combines OpenAI embeddings for semantic comprehension, Pinecone for effective vector storage and retrieval, and Groq (LLaMA 3) for response generation, all of which are managed by LangChain. Automated text extraction, dynamic chunking for effective document processing, and semantic search to obtain contextually relevant responses are all components of our methodology. Furthermore, features that improve security and personalization include file uploads, user authentication, and conversation history logging.

In order to improve information retrieval efficiency, this research combines RAG, semantic search, and LLMs to advance AI-powered document analysis. It can be used in a variety of fields, including legal research, healthcare, and education, where precise document analysis is necessary for making decisions.

Motivation

In fields like education, research, and business, where conventional search techniques frequently yield inaccurate results, our Multi-PDF Question-Answering System uses AI technologies such as LLMs, semantic search, and RAG to improve information extraction. By automating text processing and providing context-aware responses, the system enhances efficiency and decision-making in a variety of fields.

Literature Survey

Resource Paper Name	Description	Limitations
Multi-PDF Question-Answering Systems	Provides context-aware answers using RAG and LLMs.	Struggles with scalability, accuracy, and user management.
Semantic Search for Document Retrieval	Uses Transformer models for semantic search to retrieve documents	Lacks user authentication, file uploads, and chat history logging.
AI-Powered Chatbots for Document Analysis	Analyzes documents using NLP and machine learning	Does not address data security, computational demands, or ethical concerns.

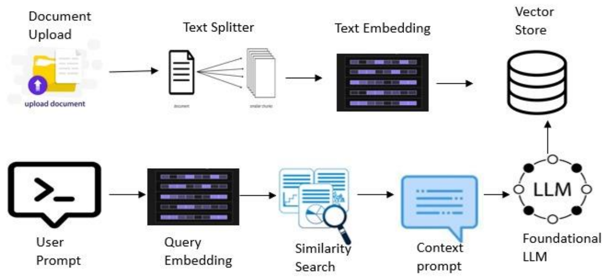
Proposed System

The proposed AI-powered document processing system is a full-featured solution that allows users to upload, store, and interact with documents using natural language queries. Built on a Flask backend with MongoDB for data storage, the system includes safe user authentication with bcrypt password hashing, allowing users to register and log in to access document functionality. The document processing pipeline accepts different file types (PDFs and text files) and automatically extracts and chunks text before building embeddings with OpenAI's text-embedding-3-large model and storing them in Pinecone for efficient semantic search. Users can communicate with their documents via a Groq-powered LLM (Llama-3-70b), which fetches relevant document chunks and delivers context-aware responses while keeping a complete conversation history for continuity.
The system provides security through role-based access control, encrypted document storage, and metadata-based access limitations. The platform's modular architecture enables intuitive document interaction, high-accuracy semantic search, and scalable document management, making it ideal for research analysis, legal document review, enterprise knowledge bases, and academic applications.

Advantages Over previous system

- Efficient Information Retrieval
- Automated Document Processing
- User-Friendly Interaction
- Enhanced Security.
- Persistent Query History
- Scalability

Model Implementation



The provided diagram illustrates a document processing pipeline utilizing a large language model (LLM) and vector storage for efficient retrieval. The process begins with document upload, where a user submits a document to the system. The document is then passed through a text splitter, which divides it into smaller, manageable chunks. These chunks undergo text embedding, where they are transformed into vector representations and stored in a vector store for future retrieval. When a user submits a query prompt, it is converted into a query embedding, ensuring it is in a format that aligns with the stored document vectors. A similarity search is then conducted to retrieve the most relevant document chunks from the vector store. These retrieved chunks are structured into a context prompt, which is sent to the foundational LLM to generate an informed response. This entire process enables an efficient, context-aware document retrieval system, allowing users to query large text repositories with high accuracy and minimal processing time.

1. System Architecture
The AI-powered document processing system is built around core components that allow for efficient document retrieval and intelligent responses. The system operates in a structured pipeline, with documents uploaded, processed, and turned into vector embeddings. These embeddings are maintained in a vector database, which allows for efficient semantic search and retrieval when a user submits a query. A Large Language Model (LLM) processes the obtained document chunks to produce accurate, context-aware replies.

2. Data Flow and Processing Lifecycle
The data flow begins when a user uploads a document through a web interface. The document is text divided and embedded using models such as OpenAI embeddings. These embeddings are saved in Pinecone for quick similarity searches. When a user enters a query, it is also embedded and matched against stored vectors. The most relevant chunks are extracted and routed to the LLM via LangChain, resulting in a structured response. This allows for accurate and intelligent document processing.

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3. Semantic Search and Query Handling

The system uses semantic search to find the most contextually relevant information. Instead of keyword-based searches, embeddings capture the meaning of text, increasing retrieval accuracy. When a user submits a query, the system searches the stored vector database for the most similar document sections. These chunks, together with the user's query, are fed into the LLM to provide contextualised replies.

5. Performance Optimization

To improve system speed, asynchronous processing is introduced for backend efficiency utilising Flask. In Pinecone, the vector search is optimised using approximate nearest neighbours (ANN) techniques, which reduces search latency. Caching methods have been implemented to speed up frequent queries.

6. Testing and Validation

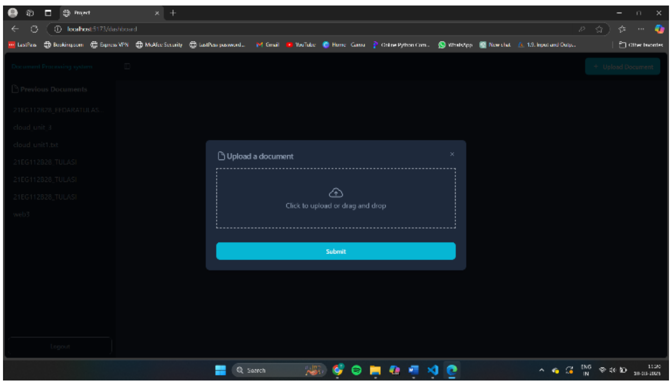
Extensive testing is conducted to validate system reliability. Load testing is performed to measure query response time and retrieval accuracy under high traffic. Security validation includes penetration testing to ensure secure document storage and data integrity. The system is assessed for accuracy, scalability, and fault tolerance, ensuring a robust AI-powered document processing solution.

Tools used

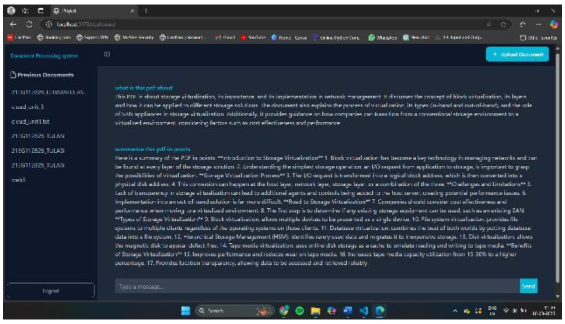
Our AI-powered document processing solution employs React.js for the user interface and Flask for backend operations. LangChain aids in the management of AI responses, whereas Pinecone enables quick and accurate document searches. MongoDB is used to securely store data. The system ensures efficient document processing and retrieval with advanced indexing methods.

Results

The upload page has interface with a drag-and-drop file upload box displaying "Click to upload or drag and drop". An upload icon indicates functionality, and a submit button allows users to confirm and upload files.



The Result Page displays the document summary and key points for quick understanding. A previous documents panel allows easy access to past uploads, while a query box lets users extract specific details dynamically.



Conclusion

Our AI-powered document processing solution utilises LangChain and semantic search to automate text extraction, summarisation, and retrieval. It combines a simple React.js UI with a Flask backend, ensuring safe MongoDB storage. This technology improves productivity, lowers manual effort, and gives correct insights, making document handling faster and more reliable for organisations working with enormous datasets.

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